



Chaitanya Priya

Senior Embedded Software Engineer | MATLAB/Simulink & Model-Based Development

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PROFESSIONAL SUMMARY

Model-based software developer with over 10 years of experience in MATLAB/Simulink development, automated code generation, and safety-critical embedded systems for the automotive sector. Delivered production Simulink models and AUTOSAR-compliant generated code for an award-winning Electric Drive Unit powering the Mercedes-AMG GT 63 S E PERFORMANCE PHEV, achieving 87% model test coverage. Experienced in Battery Management System (BMS) interface design for EV/hybrid powertrain ECUs — integrating battery State of Charge (SoC), current limits, and cell protection threshold signals via CAN FD. Certified ISO 21434 Automotive Cybersecurity Practitioner (TÜV SÜD) and ISO 26262 Functional Safety Level 1 (Omnex). Proven in ASPICE-compliant development lifecycles across Tier-1 OEM environments including Vector Informatik, American Axle & Manufacturing, and Mercedes-Benz.

CORE COMPETENCIES

Tools & Test	MATLAB & Simulink, Embedded Coder, Stateflow, Fixed-Point Toolbox, Simulink Requirements, Simulink Test, Vector CANoe/CANalyzer/CANape, VectorCast, vTESTstudio, Polyspace BF/CP, Lauterbach TRACE32, ASAP2
Model-Based Dev	Model-Based Software Development (MBD), Automated Code Generation, MIL / SIL / HIL Validation, MISRA-Compliant Generated Code, Requirements Traceability (DOORS–Simulink), AUTOSAR SWC Code Generation
Battery & EV Systems	BMS Interface Design (CAN-based), Battery State of Charge (SoC), Current Limits & Cell Protection Thresholds, P3 PHEV Powertrain, Electric Drive Unit (EDU), High-Voltage Safety, EV Charging Sequences
Programming	Embedded C, C++, Python, MATLAB scripting, Structured Text
Protocols & Interfaces	CAN, CAN FD, UDS (ISO 14229), SAE J1939, I2C, SPI, UART, MQTT, ModBus RTU/ASCII, PROFINET, RS-485
Microcontrollers & RTOS	Aurix TCx, STM32, Renesas (16/32-bit), C8051F, FreeRTOS, JDOS, RTA-OS, Bare-Metal
Automotive Standards	AUTOSAR (BSW, MCAL, DaVinci Configurator, ETAS ISOLAR-A/B, RTA-BSW), ISO 26262, ISO 21434, ASPICE (SWE.1–SWE.5)
Methodology & Process	MIL, SIL, HIL, CI/CD, MISRA, 8D, FMEA, DFMEA, IBM DOORS NG, IBM RTC, GIT, JIRA, Confluence, OpenOCD, JTAG
Design & PLM	KiCAD, Altium Designer, PreeVision, GIT, JIRA, Confluence
AI & Data	RAG Pipelines, LLM Integration, Production-Grade AI Services, Data Pipelines, Model Evaluation & Observability, Docker, Kubernetes, Generative AI (IIT Certified)

PROFESSIONAL EXPERIENCE

Senior Technical Lead

— Stuttgart, Germany

Oct 2024 – Present

Mercedes-Benz GmbH (via Soorce Professional GmbH)

Project: OnOff for IVI Infotainment ECU

Core Focus: Power State Management (PSM) **Key Tech:** Matlab/Simulink, DLT viewer, PlantUML, Doors NG, Aurix TCx, AUTOSAR, QNX, Linux, Hypervisor, AAOS

Project description:

Owned end-to-end PSM startup sequencing for safety-relevant power-state transitions. Implemented deterministic handover across Aurix, QNX, and Linux via hypervisor in a three-OS architecture. Aligned startup and shutdown logic with ASPICE SWE.1 to SWE.4 work-product expectations. Partnered with architects to harden the “OnOff” concept for ASIL-relevant scenarios and edge cases. Reduced recurring startup defects through structured debugging, trace analysis, and lifecycle issue closure.

Individual contribution :

- Developed and maintained the **Power State Manager (PSM)** for Aurix TCx-based boards and multi-OS SoCs (**AUTOSAR, QNX**, Linux via hypervisor), ensuring deterministic power state transitions in a safety-critical architecture.
- Leading migration of PSM from Linux to **Android Automotive OS (AAOS)**, re-engineering power policy alignment across a three-OS architecture (Aurix/AUTOSAR, QNX, AAOS).
- Engineered multi-stage hardware initialization logic for seamless control handoff between Aurix, QNX, and Linux.
- Collaborated with the Architecture Team to define and refine the safety-critical “OnOff” system for ASIL-relevant power states.
- Resolved complex hardware-software interface defects across the full bug resolution lifecycle, reducing recurring fault classes.
- Delivered work products aligned with ASPICE SWE.1 through SWE.4.
- Engineered Python AI microservices (RAG/LLM) for automated defect analysis and Jira triage, deployed in enterprise production via Docker/Kubernetes.

Senior Technical Lead

— Stuttgart, Germany

Jul 2023 – Sep 2024

Vector Informatik GmbH (via MSK Engineering and IT Services GmbH)

Project: EVCC for Electric Trucks

Core Focus: Charging Control Software & Compliance **Key Tech:** AUTOSAR MCAL/BSW, IBM DOORS NG, CAN, Python, ISO 26262/21434

Project description:

Designed EV charging control software with AUTOSAR MCAL/BSW integration. Implemented robust low-level driver architecture for charger ECU interfaces and boundary-condition handling. Translated BMS constraints (SoC, current limits, protection thresholds) into safe charging behavior. Strengthened SWE.4/SWE.5 verification with repeatable test evidence and static-analysis closure. Improved audit readiness for ASPICE and ISO 26262/21434 compliance reviews.

Individual contribution :

- Led requirements elicitation in **IBM DOORS NG** and authored comprehensive software design documentation for EV charging sequences, delivering ASPICE SWE.1-compliant work products.
- Analysed **Battery Management System (BMS)** interface specifications defining SoC-dependent charge limits, cell voltage protection thresholds, and balancing status communicated via CAN; ensured EV charging software compliance with all battery management boundary conditions.
- Engineered complex **device drivers** and designed robust low-level software architectures for EV charging ECUs.
- Integrated third-party libraries while maintaining CI/CD pipelines for continuous, reproducible delivery.

- Configured and maintained **AUTOSAR** base software to ensure optimal system performance.
- Guided partner teams and global customers from proof-of-concept (PoC) to scalable production rollout, coordinating vendor integrations.
- Developed rigorous test and safety plans ensuring full compliance with **ISO 26262** standards.
- Executed comprehensive unit testing, Python-based test automation, static code analysis, SIL, and integration testing as Feature Owner.
- Represented the team in ASPICE Quality and **ISO 21434** Cybersecurity Audits to ensure compliance.

Lead Embedded Software Engineer

— Detroit, USA

Oct 2019 – Jul 2023

American Axle & Manufacturing (AAM)

Project: Electric Drive Unit

Core Focus: Motor Control & Model-Based Dev (MBD)

Key Tech: MATLAB/Simulink, Stateflow, Embedded Coder, Aurix TCx, ISO 14229

Project description :

Delivered production-grade motor-control software for a P3 hybrid EDU. Developed and validated Simulink/Stateflow logic for operating modes, shifting, and torque distribution. Integrated diagnostics and safety paths with AUTOSAR-aligned interfaces and generated C code. Improved model verification depth through MIL/SIL/HIL coverage expansion and targeted defect prevention. Supported serial-production quality with ASPICE-aligned reviews and corrective-action follow-through.

Individual contribution :

- Developed **MATLAB/Simulink** models for the Motor Control ECU of AAM's rear Electric Drive Unit (EDU) powering the **Mercedes-AMG GT 63 S E PERFORMANCE** PHEV — a 150 kW / 204 hp P3 hybrid e-Drive system and 2022 Automotive News PACE Award finalist; led a sub-team of 7 engineers across 3 major Simulink subsystems.
- Designed **Stateflow** state machines to manage motor operating modes, 2-speed electromechanical gearbox shift logic, and eLSD torque distribution state transitions.
- Implemented **Battery Management System (BMS)** interface logic in Simulink to receive and process battery State of Charge (SoC), discharge current limits, and high-voltage cell protection thresholds via **CAN FD**, ensuring motor torque output remained within battery-safe operating boundaries.
- Generated production-ready **AUTOSAR**-compliant C code from Simulink using **Embedded Coder** (automated code generation); configured AUTOSAR SWC port interfaces for integration with RTA-BSW on Infineon **Aurix TCx**.
- Enforced fixed-point arithmetic and numerical precision using **Fixed-Point Toolbox** to meet Aurix embedded target constraints; verified MISRA compliance of all generated code via Polyspace BF/CP.
- Established requirements traceability between **IBM DOORS NG** software specifications and Simulink model elements via **Simulink Requirements**, enabling ASPICE SWE.3 model-level design reviews.
- Increased model test coverage from **64% to 87%** by designing and executing Model-in-the-Loop (MIL), Software-in-the-Loop (SIL), and Hardware-in-the-Loop (HIL) test suites using MATLAB test scripts — significantly reducing production defect escapes.
- Developed Embedded C/C++ software and Python scripts for vehicle diagnostics in accordance with **ISO 14229 (UDS)**.
- Integrated third-party libraries including Infineon SafeTlib and Cycliclib for security and safety compliance.
- Configured **AUTOSAR** Base Software (BSW) and Microcontroller Abstraction Layer (MCAL); maintained Git-based CI/CD pipelines and Docker-based development environments.
- Participated in customer and internal ASPICE audits (SWE.1–SWE.5) and defined corrective actions.

Lead Engineer

— Pune, India

Sep 2017 – Oct 2019

John Deere (via HCL Technologies)

Project: Skid-Steer Loader Platform

Core Focus: Application Architecture & Hardware Migration

Key Tech: MATLAB/Simulink, Embedded Coder, Renesas 32-bit, JDOS RTOS, CI/CD

Project description :

Led migration from 16-bit hardware to a 32-bit Renesas ECU. Preserved machine behavior and safety timing while refactoring model and application-layer architecture. Implemented and stabilized CAN/SPI/I2C communication pathways for control and telemetry functions.

Individual contribution :

- Integrated features into JDOS RTOS with priority-aware scheduling for safety-critical tasks.
- Improved field reliability via 8D/DFMEA-driven root-cause elimination and preventive actions.
- Developed powertrain and transmission application software for agricultural equipment (tractors) using **MATLAB/Simulink**, applying model-based development (MBD) principles across the full V-cycle from functional requirements to validated production code.
- Migrated application layer Simulink models from 16-bit microcontrollers to the 32-bit Renesas platform, refactoring model architecture and updating data types for target compatibility.
- Generated and integrated production C/C++ code using **Embedded Coder** for model-based development; validated generated code against source models via MIL/SIL equivalence checks.
- Tested features using Model-in-the-Loop (MIL) and Software-in-the-Loop (SIL) methodologies.
- Maintained and developed CI/CD pipelines (Vehicle Payload Delivery) for smooth deployments.
- Integrated, built, and debugged code for JDOS (John Deere RTOS) based vehicle projects.
- Resolved defects and performed root cause analysis using 8D, FMEA, and DFMEA techniques.
- Conducted reviews of specifications, code, and quality reports to recommend improvements.

Embedded Systems Software Developer

— Mumbai, India

Feb 2015 – Sep 2017

Libratherm Instruments Pvt. Ltd.

Projects: Thermal Controllers Family & Helium Leak Detector Machine

Core Focus: Firmware Development & Industrial IoT

Key Tech: STM32, C/C++, RS-485, Modbus RTU, PROFINET, MQTT

Project description :

Implemented STM32/Silicon Labs firmware for industrial ramp-soak controllers. Built profile state machines with PID and Servo Start behavior for controlled thermal transitions. Implemented HMI workflows (LCD/keypad) and resilient parameter retention across power cycles. Added signal-conditioning and fault logic for thermocouple integrity and process safety interlocks. Scaled RS-485 master/slave orchestration for multi-zone and multi-device deployments.

Individual contribution :

- Developed PLC-side control logic in Structured Text and Instruction List.
- Coordinated high-pressure leak-test sequencing with deterministic machine-state transitions.

- Integrated protocol communication across Modbus RTU (PC link) and PROFINET (detector link).
- Supported SCADA-facing data/control pathways for operational monitoring and maintainability.
- Improved commissioning quality through structured diagnostics, operator feedback loops, and logic tuning.
- Developed and maintained firmware for medical devices (STM32 microcontrollers, C/C++), ensuring compliance with FDA regulatory standards.
- Designed firmware for multi-zone PID temperature controllers (6-zone ramp/soak profiles, RS-485 Master/Slave), implementing closed-loop control algorithms.
- Integrated communication protocols including MQTT and ModBus RTU/ASCII for industrial IoT applications.
- Created comprehensive certification and approval documentation for product lifecycle from concept through mass production.
- Conducted rigorous unit tests to ensure firmware reliability and system performance.

Trainer - Embedded Systems Software
— Pune, India

Aug 2012 – Jan 2015
Gencor Learning Solutions Pvt. Ltd.

Project: Microcontroller Architecture & Hardware Bring-up Programs
Core Focus: Hardware Initialization & Protocols

Key Tech: GPIO, UART, SPI, I2C, FreeRTOS, MISRA C, Git

Project description :
Teaching

- Individual contribution :**
- Created structured lessons on microcontroller architectures, memory maps, and register-level programming.
 - Conducted live sessions on “board bring-up,” where trainees learn to initialize hardware from scratch.
 - Guided students in writing drivers for GPIO, UART, SPI, I2C, ADC, and PWM without high-level libraries.
 - Prepared trainees by teaching best practices in version control (Git), MISRA C standards, and RTOS like FreeRTOS.
 - Constantly monitored trainee progress by devising new tests and lab exercises.

Education	
Embedded Systems Design	University of Pune, India
B.Sc. Electronics & Telecommunication Engineering (Distinction)	Magadh University, India

Certifications	
ISO 21434:2021 – Automotive Cybersecurity Practitioner	TÜV SÜD
ISO 26262:2018 – Functional Safety Level 1	Omnex
Certified in ML and Generative AI for Professionals	Indian Institute of Technology (IIT)

Languages	
English	Fluent (C2)
German	B1 (enrolled in B2 course)
Hindi	Native